

Literal Beta Rank-Midpoint Asymptotics and Diagonal-Dominant Adjoint Phase

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Abstract

For the literal V59 coefficient on $(x/2, x] \cap \mathbb{Z}$, we evaluate its coefficient-independent ordered-rank midpoint Haar moment. Exact layer-by-layer divisor-density cancellation removes the truncated Möbius lane with error $O(x^{-67/400})$, while the second-order curvature of Li gives

$$\langle z_{\text{mid}}, \beta \rangle = \frac{\log(32/27)}{\sqrt{2}} \frac{\sqrt{x}}{\log^2 x} + O\left(\frac{\sqrt{x}}{\log^3 x}\right) > 0.$$

Substitution into the exact TPC-255 adjoint decomposition shows that the returned diagonal dominates the input-unit, hard-window, and child-jump lanes. The two boundary terms have exponent $55/48$, strictly below the diagonal exponent $56/48$ by $1/48$. Consequently the generally complex scalar $\langle z_{\text{mid}}, A_x \beta \rangle$ has an explicit negative-real leading asymptotic, is eventually nonzero, and has normalized phase tending to -1 . We do not claim that the scalar is real, that its unqualified principal argument tends to $+\pi$, or that one Haar projection closes full Gate B.

1 Literal clock and ordered-rank geometry

Let x tend to infinity through real values and put

$$a = \lfloor x/2 \rfloor, \quad b = \lfloor x \rfloor, \quad I_x = \{a+1, \dots, b\}, \quad N = b - a. \quad (1)$$

The primary split is by ordered rank [10]:

$$\ell = \lfloor N/2 \rfloor, \quad r = N - \ell, \quad m = a + \ell, \quad L = \{a+1, \dots, m\}, \quad R = \{m+1, \dots, b\}, \quad (2)$$

$$\rho^2 = \frac{\ell r}{N}, \quad z_{\text{mid}} = \rho \left(\frac{\mathbf{1}_L}{\ell} - \frac{\mathbf{1}_R}{r} \right). \quad (3)$$

The counting inner product is conjugate-linear in its first slot: $\langle f, g \rangle = \sum_{n \in I_x} \overline{f(n)} g(n)$. Thus $\|z_{\text{mid}}\|_2 = 1$ and

$$\langle z_{\text{mid}}, f \rangle = \rho \left(\frac{1}{\ell} \sum_{n \in L} f(n) - \frac{1}{r} \sum_{n \in R} f(n) \right) \quad (4)$$

for every real-valued f .

Lemma 1 (Real-clock endpoints). *Uniformly for real x ,*

$$N = \frac{x}{2} + O(1), \quad \ell, r = \frac{x}{4} + O(1), \quad m = \frac{3x}{4} + O(1), \quad \rho = \frac{\sqrt{x}}{2\sqrt{2}} + O(x^{-1/2}).$$

Moreover

$$\rho \left(\frac{1}{\ell} + \frac{1}{r} \right) = \frac{1}{\rho}, \quad |z_{\text{mid}}(n)| \leq \rho^{-1}. \quad (5)$$

Proof. The endpoint assertions follow directly from the floor errors in (1) and the rank rounding in (2). Expanding $\rho^2 = \ell r/N = x/8 + O(1)$ proves the displayed asymptotic. Finally, $\rho(1/\ell + 1/r) = \rho N/(\ell r) = 1/\rho$. On L , for example, $(\rho/\ell)/(1/\rho) = r/N \leq 1$; the R case is symmetric. $\square$

Retain the literal V59 scales and coefficient [9, 4]:

$$H = x^{21/32}, \quad Q = x^{1/3}, \quad U = x^{133/400}, \quad \mathcal{Q}_x = \{q \text{ prime} : Q < q \leq 2Q\}, \quad (6)$$

$$\beta(t) = \frac{\Lambda(t)}{\log t} - \sum_{\substack{d|t \\ d \leq U}} \mu(d), \quad t \in I_x. \quad (7)$$

2 The literal beta Haar asymptotic

Write $P(t) = \Lambda(t)/\log t$ and $D_U(t) = \sum_{d|t, d \leq U} \mu(d)$, so that $\beta = P - D_U$. The divisor lane requires no cancellation estimate for μ .

Lemma 2 (Exact layerwise density cancellation). *One has*

$$|\langle z_{\text{mid}}, D_U \rangle| \leq \rho U \left(\frac{1}{\ell} + \frac{1}{r} \right) = \frac{U}{\rho} = O(x^{-67/400}). \quad (8)$$

Proof. If J is any interval of s consecutive integers, then $\#\{n \in J : d \mid n\} = s/d + \theta_{J,d}$ with $|\theta_{J,d}| \leq 1$. Inserting this separately on L and R into (4), the two $1/d$ densities cancel for every individual d before any triangle inequality. Hence

$$|\langle z_{\text{mid}}, D_U \rangle| \leq \rho \sum_{d \leq U} |\mu(d)| \left(\frac{1}{\ell} + \frac{1}{r} \right) \leq \rho U \left(\frac{1}{\ell} + \frac{1}{r} \right).$$

Lemma 1 and $133/400 - 1/2 = -67/400$ finish the proof. $\square$

For the prime-power lane define

$$F(y) = \sum_{2 \leq n \leq y} \frac{\Lambda(n)}{\log n}.$$

The classical de la Vallée Poussin prime number theorem [1, 3, 2], source-locked for this program in [5], states

$$\pi(y) = \text{Li}(y) + O\left(y e^{-c\sqrt{\log y}}\right). \quad (9)$$

Since

$$F(y) = \pi(y) + \sum_{k \geq 2} \frac{1}{k} \pi(y^{1/k}) = \pi(y) + O(\sqrt{y} \log y),$$

the prime-power tail is absorbed by a possibly smaller constant in (9):

$$F(y) = \text{Li}(y) + O\left(y e^{-c_1 \sqrt{\log y}}\right). \quad (10)$$

Lemma 3 (Second-order Li curvature). *The ordered children satisfy*

$$\frac{F(m) - F(a)}{\ell} - \frac{F(b) - F(m)}{r} = \frac{2 \log(32/27)}{\log^2 x} + O\left(\frac{1}{\log^3 x}\right). \quad (11)$$

Proof. The error in (10), divided by a child length comparable to x , is $O(e^{-c_2 \sqrt{\log x}})$. Replacing the integer endpoints by $x/2, 3x/4, x$ changes each normalized Li difference by $O(1/(x \log x))$. Uniformly for $y \in [1/2, 1]$,

$$\frac{1}{\log(xy)} = \frac{1}{\log x} - \frac{\log y}{\log^2 x} + O\left(\frac{1}{\log^3 x}\right).$$

Both children have scaled length $1/4$, so their $1/\log x$ terms cancel. The coefficient of $1/\log^2 x$ in the lower-minus-upper difference is

$$4 \left(\int_{3/4}^1 \log y \, dy - \int_{1/2}^{3/4} \log y \, dy \right) = 2 \log \frac{32}{27}.$$

The endpoint, normalization, and PNT errors are all absorbed in the stated remainder. $\square$

Theorem 4 (Literal beta rank-midpoint asymptotic). *For all sufficiently large real x ,*

$$\langle z_{\text{mid}}, \beta \rangle = \frac{\log(32/27)}{\sqrt{2}} \frac{\sqrt{x}}{\log^2 x} + O\left(\frac{\sqrt{x}}{\log^3 x}\right) > 0. \quad (12)$$

Proof. Multiply (11) by $\rho = \sqrt{x}/(2\sqrt{2}) + O(x^{-1/2})$ and subtract the divisor contribution from Lemma 2. Since $x^{-67/400} = o(\sqrt{x}/\log^3 x)$, this gives (12). Its main constant is positive. $\square$

3 Exact adjoint lanes and their estimates

Fix a smooth profile ψ_+ supported in $[-1, 1]$, normalized by $\int \psi_+ = 1$, and set $K_H(h) = \widehat{\psi}_+(h/H)$. No reality or evenness is assumed. The literal operator is [7, 6]

$$A_x(u, t) = \mathbf{1}_{u \neq t} \sum_{q \in \mathcal{Q}_x} q \mathbf{1}_{q \nmid u} \mathbf{1}_{q \nmid t} K_H(u - t) \left(\mathbf{1}_{u \equiv t \pmod{q}} - \frac{1}{q-1} \right). \quad (13)$$

For $q \nmid t$ define the complete combined unit row

$$v_{q,t}(u) = \mathbf{1}_{q \nmid u} \left(\mathbf{1}_{u \equiv t \pmod{q}} - \frac{1}{q-1} \right). \quad (14)$$

It is centered over a complete period. The TPC-255 adjoint compiler, using the band-limited complete-lattice Poisson zero from V43 [8, 6], gives for all sufficiently large x (hence $H > 2Q$)

$$S_x := \langle z_{\text{mid}}, A_x \beta \rangle = -B_Q \langle z_{\text{mid}}, \beta \rangle + R_{\text{unit}} + R_{\text{hard}} + R_{\text{jump}}, \quad (15)$$

where

$$B_Q = \sum_{q \in \mathcal{Q}_x} \frac{q(q-2)}{q-1}, \quad (16)$$

$$R_{\text{unit}} = \sum_{q \in \mathcal{Q}_x} \frac{q(q-2)}{q-1} \sum_{\substack{t \in I_x \\ q \nmid t}} z_{\text{mid}}(t) \beta(t), \quad (17)$$

$$R_{\text{hard}} = - \sum_{q \in \mathcal{Q}_x} q \sum_{\substack{t \in I_x \\ q \nmid t}} \beta(t) z_{\text{mid}}(t) \sum_{u \notin I_x} K_H(u-t) v_{q,t}(u), \quad (18)$$

$$R_{\text{jump}} = \sum_{q \in \mathcal{Q}_x} q \sum_{\substack{t \in I_x \\ q \nmid t}} \beta(t) \sum_{u \in I_x} K_H(u-t) v_{q,t}(u) (z_{\text{mid}}(u) - z_{\text{mid}}(t)). \quad (19)$$

The adjoint identity behind (15) is $\langle z_{\text{mid}}, A_x \beta \rangle = \langle A_x^* z_{\text{mid}}, \beta \rangle$ because the first slot is conjugate-linear. Conjugating $A_x^* z_{\text{mid}}$ restores K_H in (18)–(19); no self-adjointness is used.

Lemma 5 (Weighted diagonal coefficient).

$$B_Q = \left(\frac{9}{2} + o(1) \right) \frac{x^{2/3}}{\log x}. \quad (20)$$

Proof. The exact identity $q(q-2)/(q-1) = q-1-1/(q-1)$ and weighted PNT give

$$B_Q = \sum_{Q < q \leq 2Q} q + o(Q^2/\log Q) = \left(\frac{3}{2} + o(1) \right) \frac{Q^2}{\log Q}.$$

This frozen weighted-prime input is also recorded in [11]. Since $Q = x^{1/3}$, (20) follows. $\square$

Lemma 6 (Combined unit-mask first moment). *For $q \in \mathcal{Q}_x$, $q \nmid t$, and $h \in \mathbb{Z}$,*

$$|v_{q,t}(t+h)| \leq \mathbf{1}_{q|h} + \frac{2}{q}. \quad (21)$$

Moreover, uniformly for $q \leq 2Q < H$,

$$\sum_{h \in \mathbb{Z}} |hK_H(h)| \left(\mathbf{1}_{q|h} + \frac{2}{q} \right) \ll_{\psi} \frac{H^2}{q}. \quad (22)$$

Proof. If $q \mid (t+h)$, the left side of (21) is zero. Otherwise, if $q \mid h$ it is $(q-2)/(q-1) \leq 1$, and if $q \nmid h$ it is $1/(q-1) \leq 2/q$. This proves the pointwise bound while retaining the full output-unit mask.

For any fixed $A > 3$, Schwartz decay gives $|K_H(h)| \ll_{A,\psi} (1+|h|/H)^{-A}$. Therefore

$$\sum_{q|h} |hK_H(h)| = q \sum_{k \in \mathbb{Z}} |kK_H(qk)| \ll_{\psi} H^2/q,$$

because $H/q > 1$; also $q^{-1} \sum_h |hK_H(h)| \ll_{\psi} H^2/q$. Combining the two estimates proves (22). This is the same source-backed centered first-moment mechanism as in [8]. $\square$

Proposition 7 (Boundary power separation). *For every fixed $\varepsilon > 0$,*

$$R_{\text{unit}} = O_{\varepsilon}(x^{5/6+\varepsilon}), \quad (23)$$

$$R_{\text{hard}}, R_{\text{jump}} = O_{\psi,\varepsilon}(x^{55/48+\varepsilon}). \quad (24)$$

Proof. The divisor bound gives $|\beta(t)| \leq 1 + \tau(t) \ll_{\varepsilon} x^{\varepsilon}$ on I_x . Using $|z_{\text{mid}}(t)| \leq \rho^{-1}$ and at most $x/q + 1$ multiples of q ,

$$|R_{\text{unit}}| \ll \frac{x^{\varepsilon}}{\rho} \sum_{q \in \mathbb{Q}_x} q \left(\frac{x}{q} + 1 \right) \ll \frac{x^{1+\varepsilon} Q}{\rho} = O(x^{5/6+\varepsilon}).$$

For each fixed displacement h , at most $|h|$ values of $t \in I_x$ have $t+h \notin I_x$. Apply (21) and (22) directly to the combined row, without splitting its zero modes. This gives

$$|R_{\text{hard}}| \ll_{\psi,\varepsilon} \frac{x^{\varepsilon}}{\rho} \sum_{q \in \mathbb{Q}_x} q \frac{H^2}{q} \ll_{\psi,\varepsilon} \frac{QH^2}{\rho} x^{\varepsilon}.$$

For R_{jump} , the difference $z_{\text{mid}}(u) - z_{\text{mid}}(t)$ vanishes inside either child and has absolute value $1/\rho$ across the midpoint by (5). At a fixed h , the number of pairs crossing that single boundary is again at most $|h|$. The identical first-moment estimate therefore applies. Finally

$$\frac{QH^2}{\rho} = x^{1/3+2(21/32)-1/2+o(1)} = x^{55/48+o(1)},$$

which proves (24) after absorbing the harmless divisor envelope into x^{ε} . $\square$

The combined row in (14) is essential. If it is written as

$$c_{q,t}(u) = \mathbf{1}_{u \equiv t(q)} - \frac{1}{q-1}, \quad d_q(u) = \frac{\mathbf{1}_{q|u}}{q-1},$$

then $v_{q,t} = c_{q,t} + d_q$, but their period sums are respectively $-1/(q-1)$ and $+1/(q-1)$. Only the sum is centered. Thus applying Poisson to either piece separately is refuted; Proposition 7 uses the combined mask throughout.

Theorem 8 (Diagonal-dominant adjoint phase). *As real $x \rightarrow \infty$,*

$$\langle z_{\text{mid}}, A_x \beta \rangle = - \left(\frac{9 \log(32/27)}{2\sqrt{2}} + o(1) \right) \frac{x^{7/6}}{\log^3 x} \quad \text{in } \mathbb{C}. \quad (25)$$

Consequently, for all sufficiently large x ,

$$\Re \langle z_{\text{mid}}, A_x \beta \rangle < 0, \quad \langle z_{\text{mid}}, A_x \beta \rangle \neq 0, \quad \frac{\langle z_{\text{mid}}, A_x \beta \rangle}{|\langle z_{\text{mid}}, A_x \beta \rangle|} \longrightarrow -1. \quad (26)$$

Proof. Combine Theorem 4, Lemma 5, and the exact identity (15). The diagonal has exponent $2/3+1/2 = 7/6 = 56/48$. Choose any fixed $0 < \varepsilon < 1/48$ in Proposition 7; the boundary exponent is $55/48 + \varepsilon$, so the gap is

$$\frac{56}{48} - \frac{55}{48} = \frac{1}{48},$$

and even the factor $\log^3 x$ is absorbed by this fixed power. The input-unit term is smaller still. This proves (25) as a complex asymptotic. Division by its positive real scale gives a strictly negative real limit, which implies all three statements in (26). $\square$

Remark 9 (Phase firewall). *The remainder in (25) may be nonreal because K_H need not be real or even. The theorem does not say that the scalar itself is real. It also does not force the unqualified principal argument to approach $+\pi$: approach from opposite half-planes may select opposite endpoints of the principal branch. The invariant statement is the normalized phase in (26), equivalently convergence to π modulo 2π .*

4 Validation, status, and remaining gate

The executable artifact verifies source hashes at the frozen TPC-255 release, 64 exact rank clocks, 1,536 divisor layers, 8,814 full unit-mask terms, and both boundary-counting rules. An independent implementation imports no producer code and rejects 112 deterministic semantic mutations in 14 classes. A separate suite checks 192 exact families, evenly split between integer and noninteger clocks. The finite values

$$0.126722052900396 \quad (x = 10^5), \quad 0.119396549784227 \quad (x = 10^6)$$

are scaled beta-Haar *numerical observations*; they receive no proof credit. The proved target constant is $\log(32/27)/\sqrt{2} = 0.120136761035088 \dots$

The exact route classification is a scoped arithmetic advance:

Marker	Value
TPC256_ROUTE_ADVANCE	YES_LITERAL_ARITHMETIC
TPC256_ARITHMETIC_ADVANCE	YES_SCOPED_LITERAL_BETA_ADJOINT_HAAR_LANE
TPC256_LITERAL_BETA_HAAR_ASYMPTOTIC	PROVED_SOURCE_BACKED
TPC256_ADJOINT_NORMALIZED_COMPLEX_ASYMPTOTIC	PROVED_SOURCE_BACKED
TPC256_REAL_PART_EVENTUALLY_NEGATIVE	PROVED
TPC256_SCALAR_EVENTUALLY_NONZERO	PROVED
TPC256_NORMALIZED_PHASE_TO_MINUS_ONE	PROVED
TPC256_SCALAR_IS_REAL	NOT_CLAIMED
TPC256_UNQUALIFIED_PRINCIPAL_ARGUMENT_TO_PLUS_PI	NOT_CLAIMED
TPC256_FIXED_ATOM_CREDIT	0
TPC256_L2	NONE
TPC256_FULL_GATE_B	OPEN
TPC256_FULL_GATE_B_STRICT_1_OVER_400	UNPAID_GLOBAL
TPC256_TWIN_PRIME_RESULT	NONE

Strongest result and obstruction. The literal β midpoint moment has an explicit positive asymptotic, and the deleted-diagonal B_Q return forces an explicit negative-real leading asymptotic for the literal adjoint Haar scalar. The corresponding obstruction is structural: the Poisson zero does not make this lane small; diagonal deletion returns the asymptotically dominant term. Moreover, the two output-unit pieces cannot be separated before exact recentering.

Open theorem and reusable structure. One ordered-rank Haar projection does not control the transverse/full-output component of $A_x\beta$, couple it to the physical w lane, prove an L^2 estimate, or pay full Gate B. The reusable chain is

layerwise divisor density cancellation $\longrightarrow$ second-order PNT curvature
 $\longrightarrow$ literal beta Haar main term $\longrightarrow B_Q$ amplification
 $\longrightarrow$ first-moment boundary separation.

The retained next clue is

EXPLOIT_EXACT_DIVISOR_DENSITY_CANCELLATION_BEFORE_ANY_TRIANGLE__THEN_USE_THE_BQ_DIAGONAL_MAIN_AND_H2_OVER_Q_BOUNDARY_MOMENT_TO_ISOLATE_THE_TRANSVERSE_FULL_GATE_B_REMAINDER.

References

- [1] Harold Davenport. *Multiplicative Number Theory*. Springer, third edition, 2000.
- [2] Henryk Iwaniec and Emmanuel Kowalski. *Analytic Number Theory*. American Mathematical Society, 2004.
- [3] Hugh L. Montgomery and Robert C. Vaughan. *Multiplicative Number Theory I: Classical Theory*. Cambridge University Press, 2007.
- [4] Liang Wang. Compensated pair dilation and angular dispersion. Prime Dynamics Theory, Bridge B V51, 2026.
- [5] Liang Wang. Critical-depth row-mass obstruction. Prime Dynamics Theory, TPC-233 source lock, 2026.
- [6] Liang Wang. Exact adjoint diagonal return and hard-boundary decomposition for the literal V59 haar lane. Prime Dynamics Theory, TPC-255, 2026.
- [7] Liang Wang. Polarized local BDH scalar compiler. Prime Dynamics Theory, Bridge B V59, 2026.
- [8] Liang Wang. Proper-factor poisson transference. Prime Dynamics Theory, Bridge B V43, 2026.
- [9] Liang Wang. Proper factors and the coprime fixed-shift ratio core. Prime Dynamics Theory, Bridge B V35, 2026.
- [10] Liang Wang. Source-frozen rank-midpoint contrasts for the literal V59 scalar. Prime Dynamics Theory, TPC-253, 2026.
- [11] Liang Wang. Top-prime direct energy floor. Prime Dynamics Theory, Bridge B internal report, 2026.